Please complete all of the following problems.
Show all your work.Consult the textbook if you need more information.

7.2 Consider the following types of electromagnetic radiation:
(1) microwave (2) ultraviolet (3) radio waves
(4) infrared (5) x-ray (6) visible
(a) Arrange them in order of increasing wavelength.
(b) Arrange them in order of increasing frequency.
(c) Arrange them in order of increasing energy.

7.7 An AM station broadcasts rock music at "960 on your radio dial." Units for AM frequencies are given in kilohertz (kHz). Find the wavelength of the station's radio waves in meters (m), nanometers (nm), and angstroms (Å).

(find how many angstroms are in a meter)

7.9 A radio wave has a frequency of 3.6×10^{10} Hz. What is the energy (in J) of one photon of this radiation?

7.11 Rank the following photons in terms of increasing energy: (a) blue ($\lambda = 453$ nm); (b) red ($\lambda = 660$ nm); (c) yellow ($\lambda = 595$ nm).

7.13 Police often monitor traffic with "K-band" radar guns, which operate in the microwave region at 22.235 GHz (1 GHz $= 10^9$ Hz). Find the wavelength (in nm and Å) of this radiation.

7.16 (a) The first step in the formation of ozone in the upper atmosphere occurs when oxygen molecules absorb UV radiation of wavelengths ≤ 242 nm. Calculate the frequency and energy of the least energetic of these photons.
(b) Ozone absorbs light having wavelengths of 2200 to 2900 Å, thus protecting organisms on Earth's surface from this high-energy UV radiation. What are the frequency and energy of the most energetic of these photons?

7.20 Which of these electron transitions correspond to absorption of energy and which to emission?
(a) $n = 2$ to $n = 4$ (b) $n = 3$ to $n = 1$
(c) $n = 5$ to $n = 2$ (d) $n = 3$ to $n = 4$

7.22 The H atom and the Be^{3+} ion each have one electron. Does the Bohr model predict their spectra accurately? Would you expect their line spectra to be identical? Explain.

7.23 Use the Rydberg equation (Equation 7.3) to calculate the wavelength (in nm) of the photon emitted when a hydrogen atom undergoes a transition from $n = 5$ to $n = 2$.

7.25 What is the wavelength (in nm) of the least energetic spectral line in the infrared series of the H atom?

7.31 The electron in a ground-state H atom absorbs a photon of wavelength 97.20 nm. To what energy level does the electron move?

7.39 A 220-lb fullback runs the 40-yd dash at a speed of 19.6 ± 0.1 mi/h.
(a) What is his de Broglie wavelength (in meters)?
(b) What is the uncertainty in his position?

7.43 A sodium flame has a characteristic yellow color due to emissions of wavelength 589 nm. What is the mass equivalence of one photon of this wavelength ($1 \text{ J} = 1 \text{ kg·m}^2/\text{s}^2$)?

7.47 Explain in your own words what it means for the peak in the radial probability distribution plot for the $n = 1$ level of a hydrogen atom to be at 0.529 Å. Is the probability of finding an electron at 0.529 Å from the nucleus greater for the 1s or the 2s orbital?

7.48 What feature of an orbital is related to each of the following quantum numbers?
(a) Principal quantum number (n)
(b) Angular momentum quantum number (l)
(c) Magnetic quantum number (m_l)

7.49 How many orbitals in an atom can have each of the following designations: (a) 1s; (b) 4d; (c) 3p; (d) $n = 3$?

7.50 How many orbitals in an atom can have each of the following designations: (a) 5f; (b) 4p; (c) 5d; (d) $n = 2$?

7.51 Give all possible m_l values for orbitals that have each of the following: (a) $l = 2$; (b) $n = 1$; (c) $n = 4$, $l = 3$.

7.52 Give all possible m_l values for orbitals that have each of the following: (a) $l = 3$; (b) $n = 2$; (c) $n = 6$, $l = 1$.

7.57 For each of the following sublevels, give the n and l values and the number of orbitals: (a) 5s; (b) 3p; (c) 4f.

7.62 The quantum-mechanical treatment of the hydrogen atom gives the energy, E, of the electron as a function of the principal quantum number, n:

$$E = -\frac{h^2}{8\pi^2 m_e a_0^2 n^2} \qquad (n = 1, 2, 3, \ldots)$$

where h is Planck's constant, m_e is the electron mass, and a_0 is 52.92×10^{-12} m.

(a) Write the expression in the form $E = -(\text{constant})\frac{1}{n^2}$, evaluate the constant (in J), and compare it with the corresponding expression from Bohr's theory.
(b) Use the expression to find ΔE between $n = 2$ and $n = 3$.
(c) Calculate the wavelength of the photon that corresponds to this energy change. Is this photon seen in the hydrogen spectrum obtained from experiment (see Figure 7.8, p. 264)?

7.75 The following quantum number combinations are not allowed. Assuming the n and m_l values are correct, change the l value to create an allowable combination:
(a) $n = 3$; $l = 0$; $m_l = -1$ (b) $n = 3$; $l = 3$; $m_l = +1$
(c) $n = 7$; $l = 2$; $m_l = +3$ (d) $n = 4$; $l = 1$; $m_l = -2$